

PROJECT REPORT

Food Ordering Behaviour & Customer Trends: A Structured Analysis of Choices and Habits

Submitted by

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Tool Used: Tableau Public

Dataset: food_ordering_behavior_dataset.csv (50,000 records, 19 fields)

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1. Introduction

1.1 Project Overview

This project analyses food ordering behaviour and customer trends using a 50,000-record order dataset spanning six Indian cities (Mumbai, Pune, Bangalore, Delhi, Chandigarh, Hyderabad), six cuisines (Chinese, South Indian, Biryani, Fast Food, North Indian, Desserts) and four meal times (Breakfast, Lunch, Snacks, Dinner). Each record captures not only the transaction itself (order value, delivery fee, rating, discount used) but the behavioural context around it the customer's mood, hunger level, who they ordered with, and whether it was raining. The objective is to convert this combination of transactional and behavioural data into an interactive Tableau dashboard and story that helps stakeholders understand how and why customers order the way they do, rather than just what they ordered. The finished workbook is published on Tableau Public for open, browser-based access.

1.2 Purpose

The purpose of this project is to give business analysts, restaurant partners and platform teams a single, visual, easy-to-explore analytical layer over food ordering data replacing scattered spreadsheet reports with an interactive dashboard that surfaces KPIs, regional and cuisine preferences, meal-time revenue, customer ratings and delivery-time patterns. Concretely, the dashboard is meant to answer questions such as: Which cities are driving the most orders? Which cuisines and meal types are most popular, and where? Does the type of company someone orders with change what they order? How satisfied are customers, and does that vary by meal type? And how long does it typically take to place an order? Answering these with evidence rather than guesswork is the core purpose of the solution.

2. Ideation Phase

2.1 Problem Statement

Two problem statements anchor this project — one from the end customer's perspective and one from the business/restaurant-partner perspective — since the dataset supports both views.

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	a customer using a food ordering & delivery app	get a meal that matches my mood, hunger level and occasion	the app shows the same generic menu regardless of city, weather, or who I'm ordering with	ordering platforms don't personalise recommendations using behavioural context (mood, weather, company type)	frustrated and less confident that I'm choosing the right restaurant or cuisine
PS-2	a restaurant partner / platform	understand ordering patterns	order, rating and behavioural	there is no consolidated visual analytics	unable to plan staffing, offers and

	business analyst	across cities, age groups and meal times	data (mood, weather, company, discounts) is scattered across raw logs	view connecting these factors to order value and satisfaction	inventory effectively for each city and meal time
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PS-1 and PS-2 together define the two audiences the dashboard must serve: PS-1 justifies the behavioural worksheets (mood, company, weather-linked views), while PS-2 justifies the geographic and operational worksheets (city, meal-type revenue, delivery time).

2.2 Empathy Map Canvas

The persona modelled below is an urban food-delivery customer (age 18-44) drawn directly from the behavioural fields in the dataset mood, hunger level, company type, and rainy-weather conditions rather than an invented generic user, so the map's insights translate directly into dashboard requirements.

Empathy Map Canvas — Food Ordering App Customer

SAYS • "I want my food delivered hot & fast" • "Why is delivery slower on rainy days?" • "Give me discounts on weekdays too" • "I want cuisine options that match my mood, not just my location"	THINKS • Wonders if the app understands my hunger level and mood before suggesting food • Worried about hygiene & freshness for budget-priced restaurants • Unsure if ordering with friends/family changes the recommended menu
DOES • Orders mostly in the evening/night • Chooses Biryani/Chinese on weekends • Applies discount codes ~50% of the time • Orders alone on weekdays, with family/partner on weekends	FEELS • Happy/Celebrating when ordering with partner or for an occasion • Stressed during hurried work-lunch orders • Frustrated when delivery is slow in rainy weather

Key insight: ordering behaviour is driven less by a single factor and more by the intersection of mood, occasion and weather the same customer orders very differently on a stressed weekday lunch versus a celebratory weekend dinner with a partner. This motivated the "Orders by Company Type & Meal Type" and "Distribution of Delivery Time" worksheets in the final dashboard.

2.3 Brainstorming

Ideas were gathered under three groups Visualization Ideas, Behavioural Insight Ideas, and Delivery & Ops Ideas mirroring the three angles from which the dataset can be read: what happened, why it happened, and how well it was served. Each idea was then plotted on an Impact vs. Effort matrix; high-impact, low-to-medium-effort ideas (KPI dashboard, cuisine/city breakdown, meal-type revenue & rating charts) were prioritised for this release, while high-effort items such as a full real-time data feed were pushed to Future Scope.

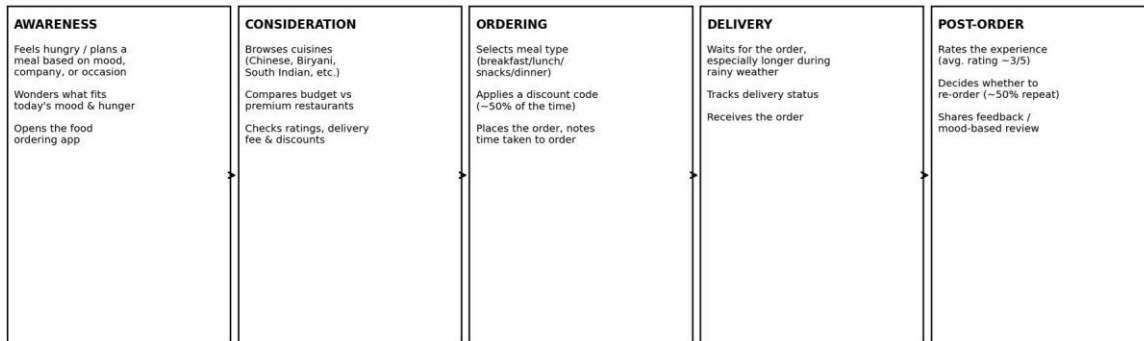


Idea Prioritization — Impact vs. Effort

3. Requirement Analysis

3.1 Customer Journey Map

Customer Journey Map — Food Ordering Behaviour & Customer Trends



3.2 Solution Requirement

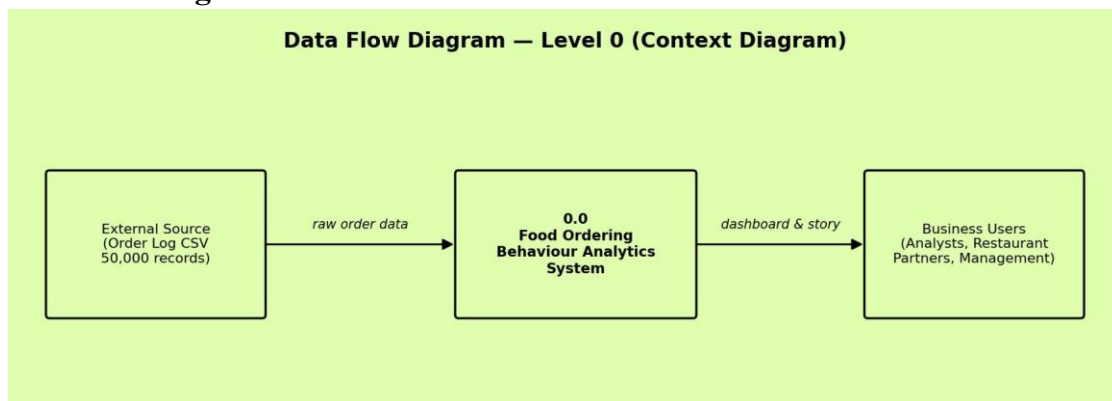
Functional Requirements:

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / SubTask)
FR-1	Data Collection & Extraction	Import food_ordering_behavior_dataset.csv (50,000 records); Validate schema 19 fields, correct data types
FR-2	Data Preparation	Handle missing/duplicate records; Create derived fields (age group, revenue, repeat-order flag)
FR-3	Visualization Development	Build worksheets for KPIs, cuisine, meal type, city, rating, delivery time; Create calculated fields
FR-4	Dashboard Assembly	Combine worksheets into a single interactive dashboard; Add quick filters and dashboard actions
FR-5	Story Creation	Sequence dashboard views into a narrative storyboard; Add captions guiding the viewer through insights
FR-6	Publishing	Publish workbook to Tableau Public; Generate and verify shareable public link

Non-Functional Requirements:

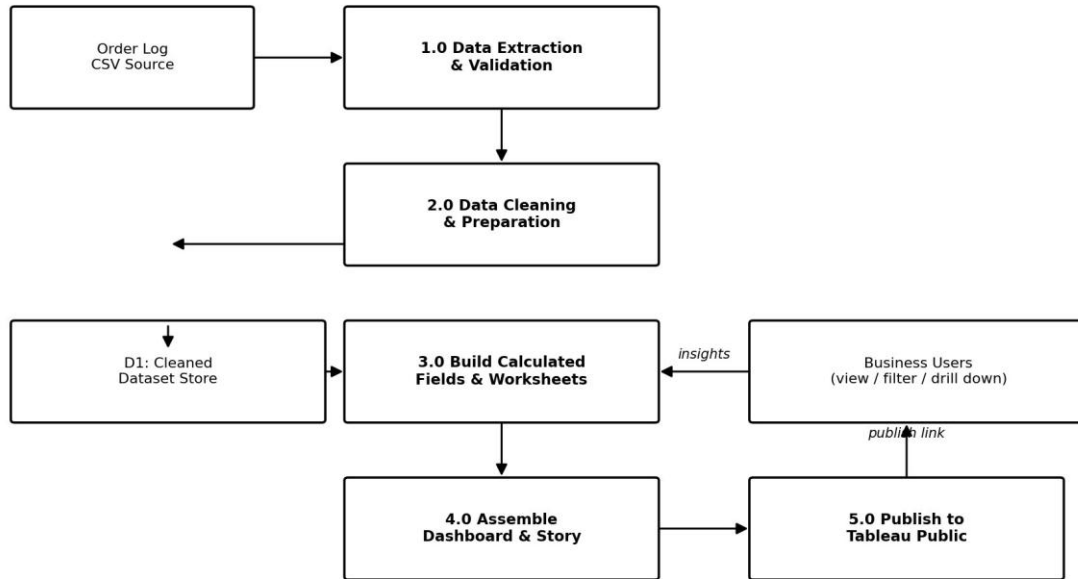
FR No.	Non-Functional Requirement	Description
NFR-1	Usability	Intuitive filters, tooltips and consistent color legends so non-technical business users can explore the dashboard without training.
NFR-2	Security	Published as a read-only public view; only anonymised user_id is exposed, no personally identifiable customer information.
NFR-3	Reliability	Dashboard produces consistent, repeatable results on every refresh of the underlying data extract.
NFR-4	Performance	Dashboard renders in under 5 seconds for the full 50,000record extract using Tableau's Hyper engine.
NFR-5	Availability	Hosted on Tableau Public, accessible 24/7 from any web browser without installation.
NFR-6	Scalability	Extract-based architecture can scale to additional cities, years, or a larger record count via incremental data refresh.

3.3 Data Flow Diagram



Level 0 — Context Diagram

Data Flow Diagram — Level 1



Level 1 — Detailed Data Flow

3.4 Technology Stack

Technology Stack — Data Flow



S.No	Component	Description	Technology
1	User Interface	Interactive Tableau Dashboard & Story in a web browser	Tableau Dashboard, Story, Tableau Public
2	Data Extraction	Reading & validating the raw order dataset	Python / Pandas, CSV
3	Data Preparation	Cleaning, type casting, derived fields	Tableau Data Prep / Excel

4	Visualization Logic	Calculated fields, worksheets, aggregations	Tableau Desktop
5	Data Engine	Fast query performance on 50,000 rows	Tableau Hyper Extract (.hyper)
6	File Storage	Workbook storage	Packaged Tableau Workbook (.twbx)
7	Infrastructure	Local build, cloud publish	Local Desktop + Tableau Public Cloud

4. Project Design

4.1 Problem – Solution Fit

Problem - Solution Fit Canvas — Food Ordering Behaviour & Customer Trends		
1. CUSTOMER SEGMENTS Urban food-delivery customers (age 18-44) across Mumbai, Pune, Bangalore, Delhi, Chandigarh, Hyderabad; restaurant partners and platform business analysts	2. JOBS-TO-BE-DONE Order food matching mood, hunger level & occasion; get it delivered quickly; for businesses — understand ordering trends to plan menus, offers & staffing	5. AVAILABLE SOLUTIONS Generic sales dashboards showing only revenue; manual spreadsheet reports; no dashboards combine mood/weather/company context with order data
3. TRIGGERS Hunger, mood swings (stressed/celebrating), rainy weather, weekend gatherings with family/friends/partner, discount notifications	7. BEHAVIOUR Order mostly evening/night; weekend orders skew Biryani & Chinese; ~50% use discounts; repeat-order rate ~50%; ratings cluster around 3/5	8. CHANNELS OF BEHAVIOUR Online: food delivery app/website Offline: word-of-mouth, restaurant signage, festive/weekend planning with family & friends
4. EMOTIONS (BEFORE / AFTER) Before: hungry, indecisive, unsure of options → After: happy/celebrating (good order) or frustrated/stressed (late delivery, wrong pick)	9. PROBLEM ROOT CAUSE Order and behavioural data (mood, weather, company, hunger) is captured but never consolidated into a single visual analytics layer for decision-making	10. YOUR SOLUTION An interactive Tableau dashboard & story analysing 50,000 orders — KPIs, city/cuisine trends, meal-type revenue, ratings & delivery time — published on Tableau Public

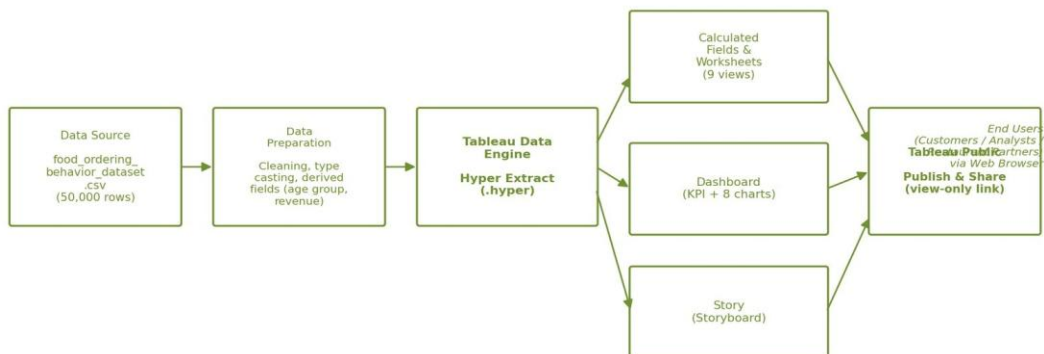
4.2 Proposed Solution

S.No.	Parameter	Description
1	Problem Statement	Businesses and customers lack a consolidated, visual view of food-ordering behaviour how mood, weather, company type and discounts influence what people order, when, and how they rate their experience across cities, cuisines and meal times.
2	Idea / Solution description	An interactive Tableau dashboard and story built on a 50,000-record order dataset, presenting KPIs, city-wise order volume, cuisine preferences, meal-type revenue, rating analysis and

		delivery-time distribution in one connected view.
3	Novelty / Uniqueness	Unlike typical revenue-only sales dashboards, this solution links behavioural context (mood, hunger level, company, rainy weather) with transactional data to explain why ordering patterns occur, not just what happened.
4	Social Impact / Customer Satisfaction	Restaurant partners can tailor menus/offers by city and meal time; platform teams can plan staffing around peak demand; customers indirectly benefit from better-targeted recommendations.
5	Business Model	Insights-as-a-service — the dashboard methodology can be licensed to restaurant chains and delivery platforms for regional strategy; the published Tableau Public version also builds analytical credibility.
6	Scalability of the Solution	Built on Tableau's extractbased (.hyper) architecture, the same workbook can scale to additional cities, longer time ranges or a larger record count without redesign.

4.3 Solution Architecture

Solution Architecture — Food Ordering Behaviour & Customer Trends



5. Project Planning & Scheduling

5.1 Project Planning

The work was organised into four sprints covering data extraction/preparation, visualization development, behavioural analysis, and story creation/publishing.

Sprint	Epic	User Story No.	Story Points			Priority
Sprint-1	Data Extraction & Preparation	USN-1, USN-2	5			High
Sprint-1	Dashboard Filtering	USN-3	2			High
Sprint-2	Dashboard Filtering / Segment Analysis	USN-4, USN-5, USN-6	7			Medium-High
Sprint-3	Rating & Delivery Insights / Behavioural Trends	USN-7, USN-8, USN-9	6			Medium
Sprint-4	Story Narrative / Publishing	USN-10, USN-11, USN-12	6			High
Sprint	Total Story Points	Duration	Start Date	End Date (Planned)	Points Completed	Release Date (Actual)
Sprint-1	7	6 Days	01 Jun 2026	06 Jun 2026	7	06 Jun 2026
Sprint-2	7	6 Days	08 Jun 2026	13 Jun 2026	7	13 Jun 2026
Sprint-3	6	6 Days	15 Jun 2026	20 Jun 2026	6	20 Jun 2026
Sprint-4	6	6 Days	22 Jun 2026	27 Jun 2026	6	27 Jun 2026

6. Functional and Performance Testing

6.1 Performance Testing

S.No.	Parameter	Screenshot / Values
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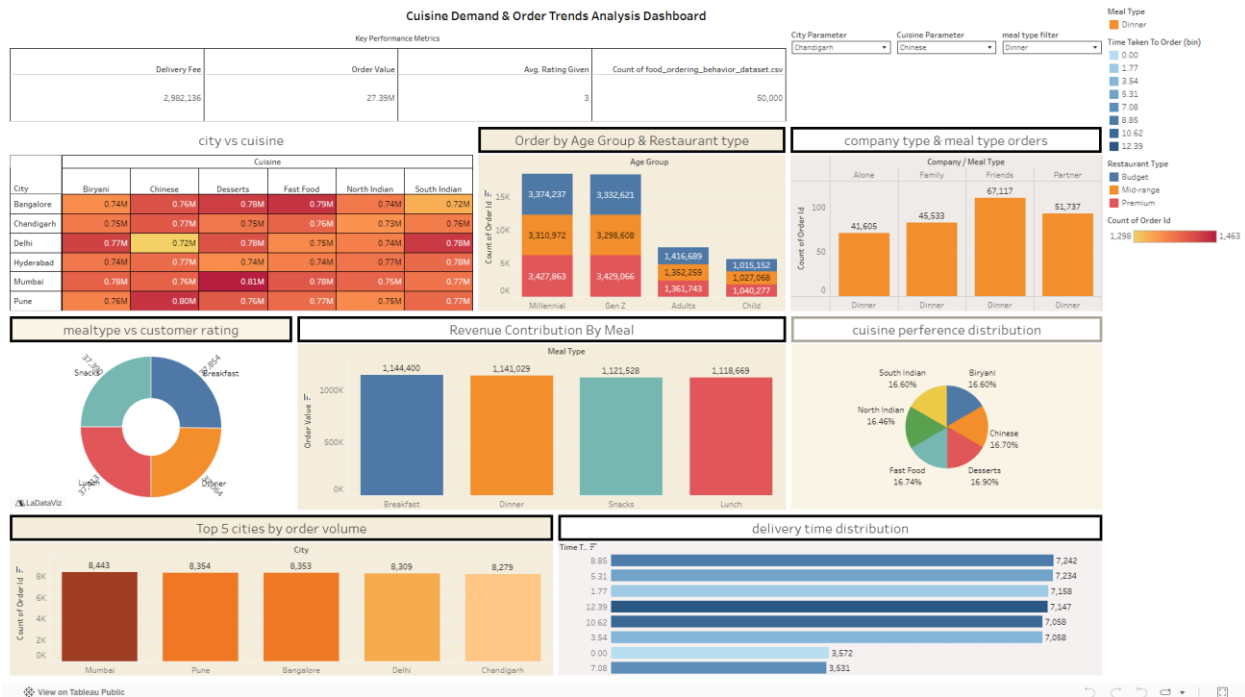
1	Data Rendered	50,000 order records across 19 fields (order_id, user_id, age, city, order_time, day_type, cuisine, meal_type, restaurant_type, order_value, discount_applied, delivery_fee, time_taken_to_order, rating_given, is_repeat_order, mood, hunger_level, company, rainy_weather)
2	Data Preprocessing	Validated schema and data types; checked for nulls/duplicates; created derived fields — Age Group, Revenue, Repeat Order Flag
3	Utilization of Filters	Quick filters on City, Cuisine, Meal Type, Day Type, Restaurant Type; dashboard actions to cross-filter KPI cards and charts
4	Calculation Fields Used	Total Revenue = SUM([order_value]); Avg Order Value = AVG([order_value]); Repeat Order Rate; Age Group (calculated bucket)
5	Dashboard Design	9 visualizations — KPI table, Flavors Across Cities, Order Distribution by Age Group, Customer Preferences Across Cuisines, Orders by Company Type & Meal Type, Customer Rating Analysis by Meal Type, Cities by Order Volume, Revenue Contribution by Meal Type, Distribution of Delivery Time
6	Story Design	1 storyboard ("food ordering behaviour") sequencing the dashboard into a business narrative

7. Results

7.1 Output Screenshots

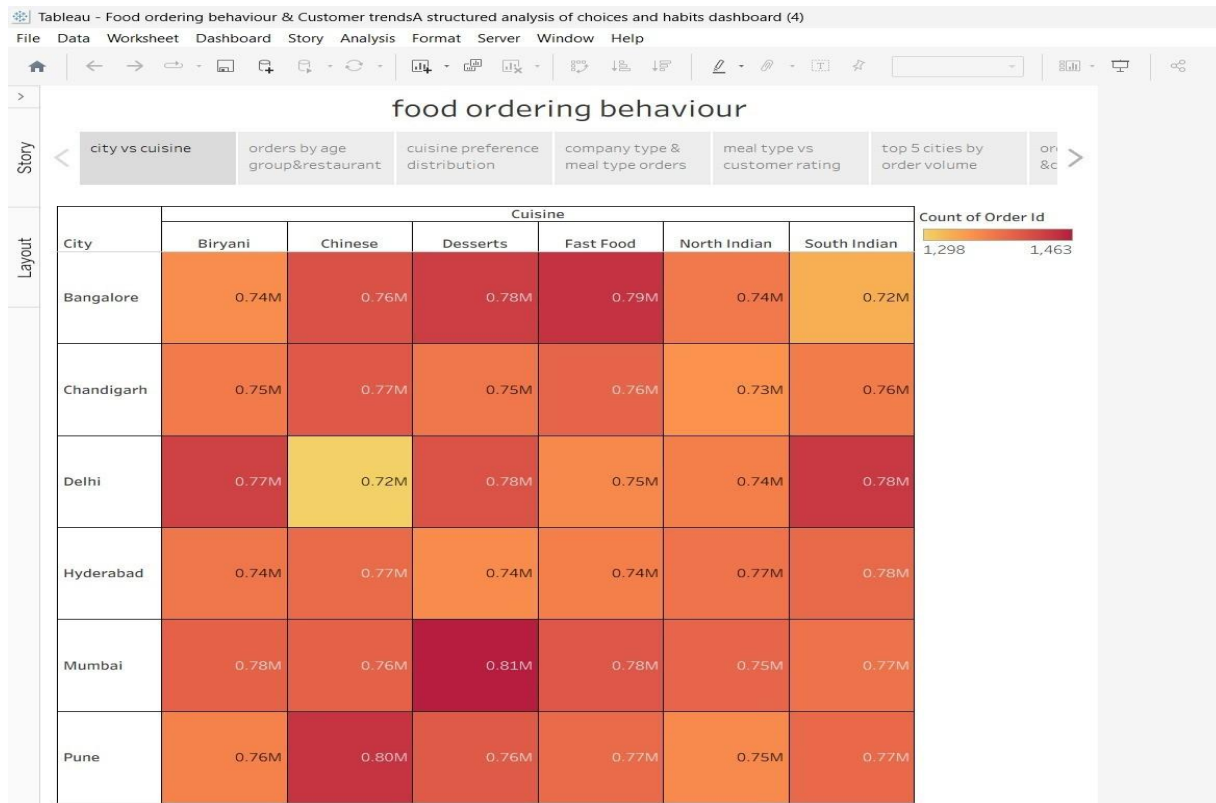
Key figures from the analysis (50,000 orders, 4,000 unique customers):

Metric	Value
Total Orders	50,000
Unique Customers	4,000
Total Revenue	₹2,73,86,555
Average Order Value	₹547.73
Average Rating	2.99 / 5
Repeat Order Rate	49.97%
Discount Usage Rate	50.23%
Most Ordered Cuisine	Desserts (16.9%), closely followed by Fast Food & Chinese (16.7%)
Highest Order Volume City	Mumbai (8,443 orders)
Highest Revenue Meal Type	Breakfast (69.2 L)₹
Total Delivery Fee Collected	₹29,82,136



Access Link:

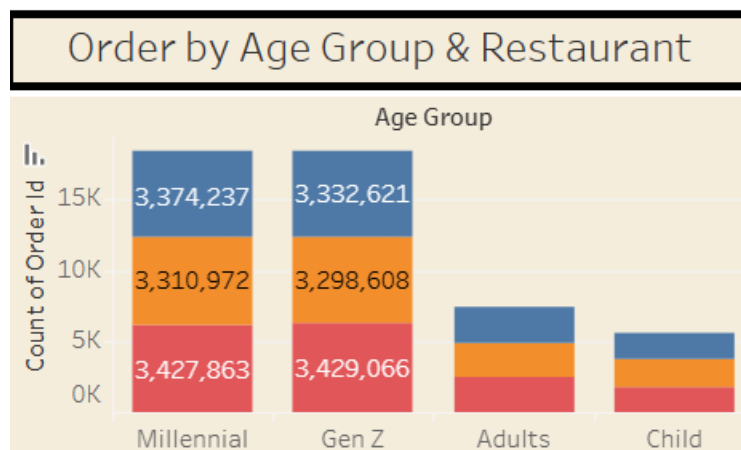
<https://public.tableau.com/app/profile/radha.reddy/viz/FoodorderingbehaviourCustomertrendsAstructuredanalysisofchoicesandhabitsdashboard4/Dashboard2?publish=yes>



Dashboard Visualization Story — "City Vs Cuisine Dashboard" (actual published Tableau dashboard)

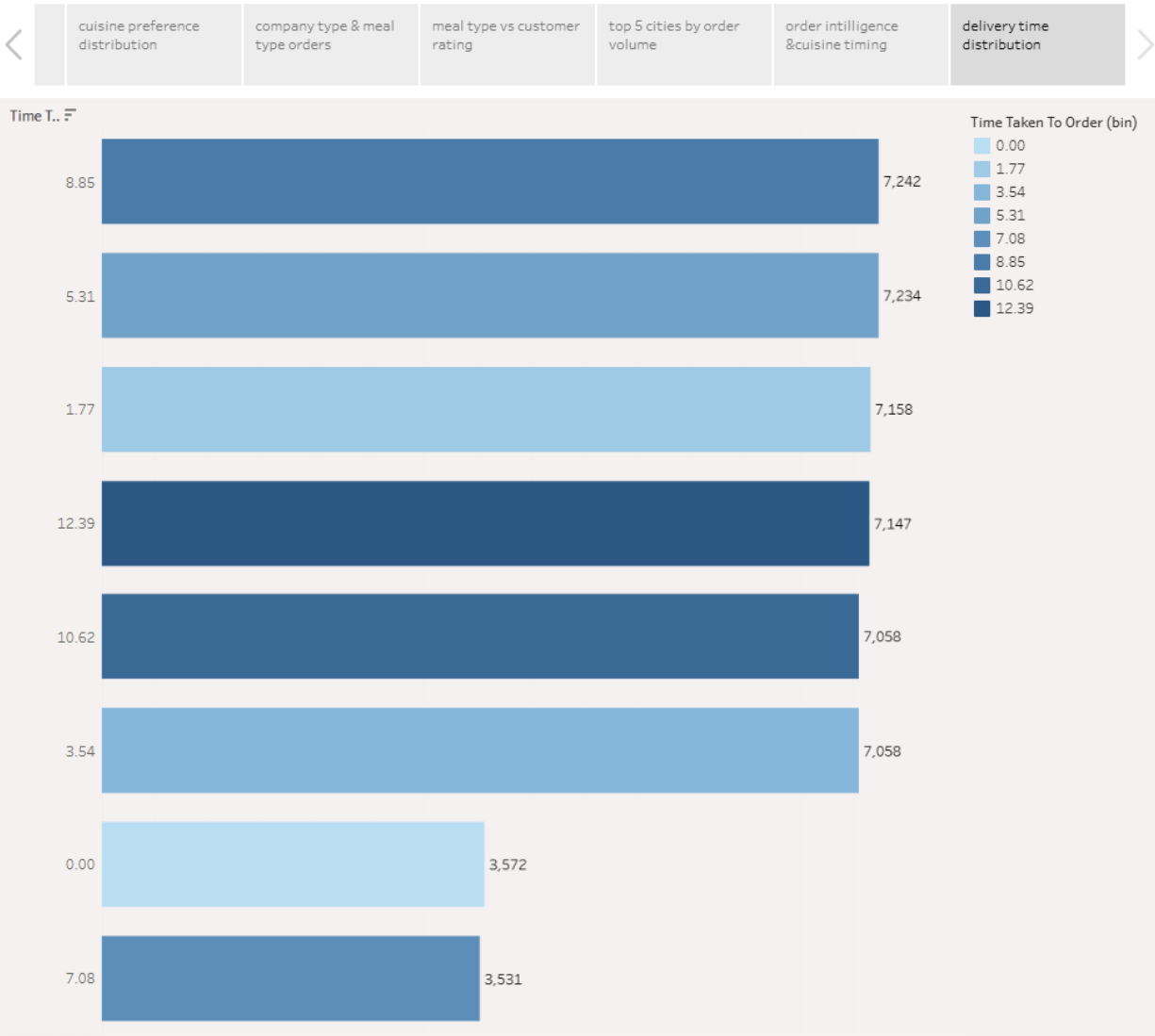
Access Link:

<https://public.tableau.com/app/profile/radha.reddy/viz/FoodorderingbehaviourCustomertrendsAstructu/redanalysisofchoicesandhabitsstory/foodorderingbehaviour?publish=yes>



The dashboard confirms the figures reported above: order demand is spread evenly across all six cities and six cuisines rather than concentrated in one segment, revenue is closely matched across the four meal types, and delivery/order time is fairly uniform across the observed range indicating a stable, predictable ordering pattern across the customer base as a whole, with behavioural factors (mood, company, weather) acting as the more meaningful differentiators explored earlier in the Empathy Map and DFD sections.

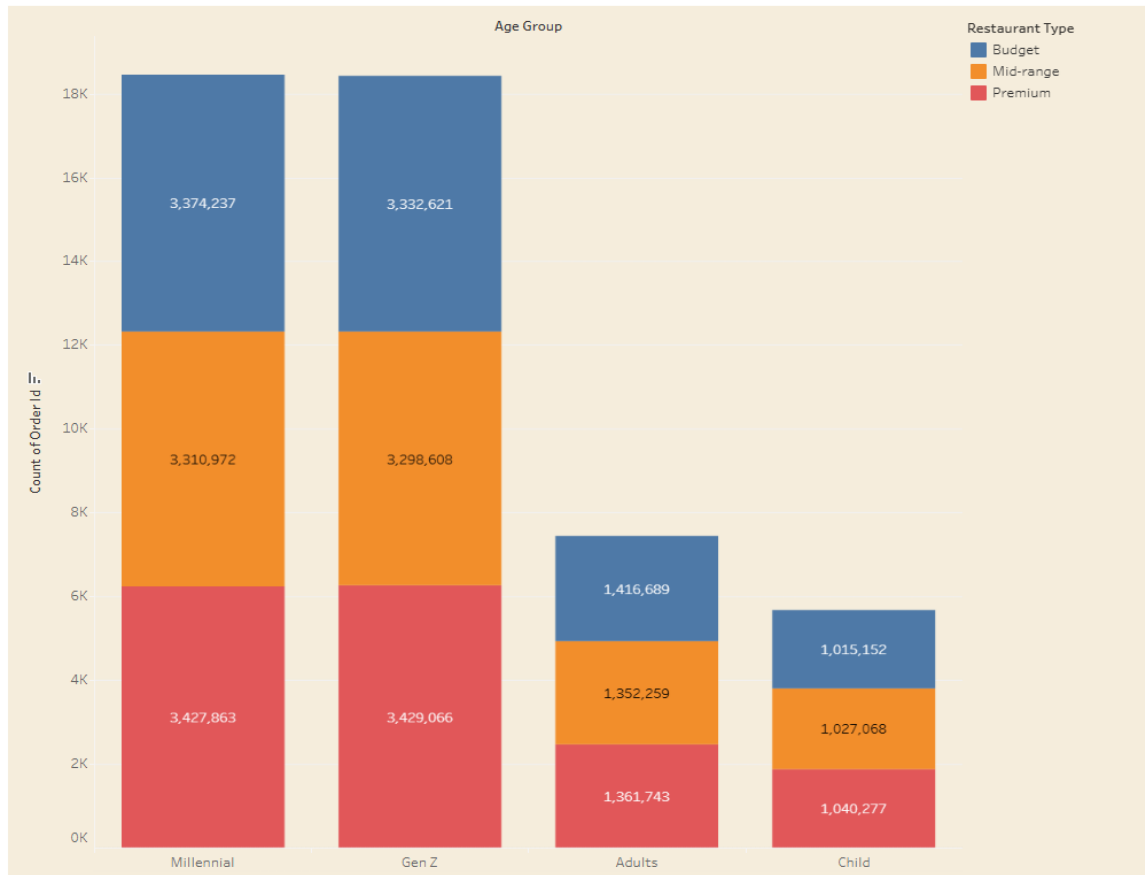
food ordering behaviour



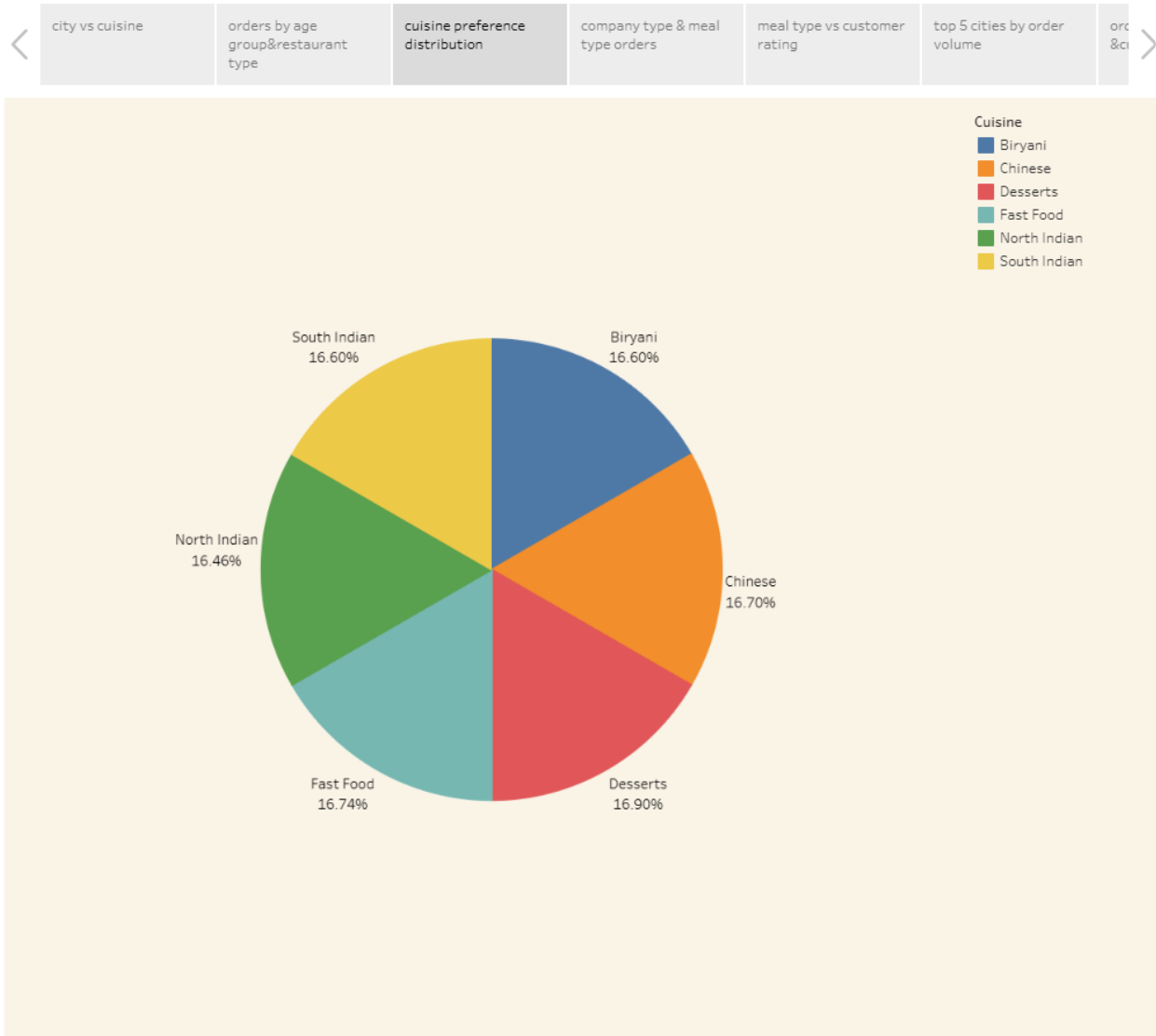
food ordering behaviour



food ordering behaviour



food ordering behaviour



Published Tableau Public Link:

<https://public.tableau.com/app/profile/radha.reddy/viz/FoodorderingbehaviourCustomertrendsAstructuredanalysisofchoicesandhabitsdashboard4/Dashboard2>

Demo Video Link:

<https://drive.google.com/file/d/1EqPvFY6cHxMu1EcVfxcAUui8lm5V3DC5/view?usp=sharing>

Github Link:

<https://github.com/radha-krishna-reddy/Food-Consumer-Trends-Analysis>

8. Advantages & Disadvantages

Advantages

- Consolidates behavioural and transactional data (mood, weather, company, discounts) into one visual layer.
- Interactive filters let non-technical stakeholders self-serve insights without writing queries.
- Extract-based architecture keeps the dashboard fast (under 5 seconds) even at 50,000 records.
- Published on Tableau Public free, shareable, accessible from any browser.
- Story view turns raw charts into a narrative that is easy to present to non-analytical stakeholders.

Disadvantages

- Tableau Public dashboards are visible to anyone with the link — not suited for confidential/proprietary data.
- No live/real-time data connection in this phase; the extract must be manually refreshed and republished.
- Limited to descriptive analytics — no predictive modelling (e.g. demand forecasting) is included yet.
- Mobile viewing on Tableau Public can compress multi-chart dashboards, reducing readability on small screens.

9. Conclusion

This project successfully transformed a 50,000-record food ordering dataset into an interactive Tableau dashboard and story. By combining transactional fields (order value, delivery fee, ratings) with behavioural context (mood, hunger level, company type, weather), the analysis gives restaurant partners and platform analysts a single view of ordering trends across six cities, six cuisines and four meal types. The dashboard's filters and calculated fields make it possible to move from a high-level KPI summary down to a specific city-cuisine-meal-type combination in a few clicks, and the published Tableau Public link makes these insights accessible to any stakeholder without needing Tableau Desktop installed.

10. Future Scope

- Connect the dashboard to a live order-management data source for real-time refresh instead of a static extract.
- Add predictive analytics (e.g. demand forecasting by city/meal-time, churn risk for repeat customers).
- Incorporate a recommendation layer that suggests cuisines based on mood, weather and company type.
- Extend the dataset to more cities and a longer time range to detect seasonal trends.
- Move from Tableau Public to Tableau Server/Cloud with row-level security if the data becomes commercially sensitive.

11. Appendix

Source Code

Data validation and pre-analysis exploration were performed using Python (pandas); the core visualization, calculated fields, dashboard and story were built natively in Tableau Desktop (no custom application source code was required for this analytics-only project).

DatasetLink:

[https://drive.google.com/file/d/1_bBUnmTq2-QOMZXj_OqOdSgijwgS8ABS/view?usp=drive link](https://drive.google.com/file/d/1_bBUnmTq2-QOMZXj_OqOdSgijwgS8ABS/view?usp=drive_link)

food_ordering_behavior_dataset.csv — 50,000 records, 19 columns (submitted alongside this report).

GitHub & Project Demo Link

Github Link:

<https://github.com/radha-krishna-reddy/Food-Consumer-Trends-Analysis>

Tableau Public Dashboard:

<https://public.tableau.com/app/profile/radha.reddy/viz/FoodorderingbehaviourCustomertrendsAstructuredanalysisofchoicesandhabitsdashboard4/Dashboard2>

Demo Link:

<https://drive.google.com/file/d/1EqPvFY6cHxMu1EcVfxCAUui8lm5V3DC5/view?usp=sharing>